



Assignment 1.1: Design World AI Platform Case Review & Engagement Orientation

Shivam Patel, Joshua Remington, Mikailou Nikiema, Mohammad Ahmed, Brian Che, and
Otaniyuwa Ero

Shiley-Marcos School of Engineering (SMSE), University of San Diego

CYBR-516-01B: Cyber Engineering Research II

Professor Douglas Magedman

06/29/2026

1. Business Context & AI Objectives

Design World (DW) is a multi-national engineering and manufacturing firm that operates in San Diego, Hong Kong, and Brussels. The leadership is preparing to deploy an enterprise level AI assistant to improve productivity while considering defensibility. The primary goals of Sentinel Cyber Consulting are to ensure all technical, operational, and legal controls are properly implemented, auditable, and align with corporate guidelines prior to launch.

2. High-Level AI Platform Description & Intended Use

The AI platform is a conversational assistant, hosted on the company's VPC on Linode, that utilizes Retrieval-Augmented Generation (RAG), agentic workflow, limited external LLM APIs, and approved tools to assist employees in their work-related duties such as document retrieval and automation.

3. Key Constraints & Non-Negotiables

The constraints and non-negotiables aspects are:

- The AI system must be exclusively deployed on the company's VPC on Linode using VMs.
- Fine-tuning or Modification of AI models is prohibited.
- The use of unvetted models or tools is prohibited.
- The use of external LLM APIs is limited.
- Governance and controls must align with NIST AI RMF, ISO/IEC 23894, OWASP Top 10 for LLMs, MITRE ATLAS, and NIST SSDF.
- The system must pass all 8 acceptance gates before launch.

4. Team Assumptions

- The VPC on Linode has already been hardened to meet cybersecurity requirements and recommendations by the previous security consulting team (CYBR-514).
- The corporate data is classified, indexed, clean, and ready to use.
- The AI platform will use only approved enterprise data sources throughout the engagement, and no additional external data sources will be introduced without formal approval.

5. Areas of Ambiguity & Uncertainty

Most of the ambiguity and uncertainty revolve around the limitations of external LLM APIs (for example, whether the limitation applies to which APIs may be used or what organizational data may be transmitted to and from the APIs), agent capabilities (such as whether the AI can write to or modify internal databases), and the project's cost cap (including whether token quotas are enforced on a daily or monthly basis and whether they apply per user or across the enterprise). It is also unclear whether different user roles, such as engineering, management,

and IT personnel, will have different levels of access to AI-generated information during the initial deployment.